New Math Commands

Integrals

Indefinite: $\int x^2$

With differential: $\int x^2 \, \mathrm{d}x$

Definite: $\int_0^1 x^2$

Definite with differential: $\int_0^1 x^2 \, \mathrm{d}x$

Contour: $\oint_C^F l$

Piecewise and systems

$$\begin{cases} x+y=3 \\ x-y=1 \end{cases}$$

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Piecewise function:

$$\begin{cases} xx>0 \\ 0x\leq 0 \end{cases}$$

Vectors and binomials

Vector: $\begin{pmatrix} 3 \\ 4 \end{pmatrix}$

Binomial: $\binom{n}{k}$

Trigonometry and hyperbolics

$\cot(x)$

$\sec(x)$

$\csc(x)$

$\arcsin(x)$

$\arccos(x)$

$\arctan(x)$

$\sinh(x)$

$\cosh(x)$

$\tanh(x)$

$\coth(x)$

Nesting, variables and calc

$\frac{\int_0^1 x}{2}$

$\int_0^2 x^2$

$\binom{3}{2}$